

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY



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FPGA-Based Automatic Washing Machine Controller

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1 Introduction

1.1 Background

Modern washing machines operate using digital controllers that automatically manage multiple sequential operations such as water filling, washing, rinsing, spinning, and draining. These processes must be executed in a structured sequence to ensure efficient cleaning and safe operation. The controller is responsible for maintaining precise timing, monitoring safety conditions, and responding to sensor inputs in real time. In embedded control systems, deterministic timing and reliability are critical factors. FPGA-based systems offer hardware-level parallelism and clock-driven deterministic behavior, making them highly suitable for time-critical applications where predictable and stable performance is required.

1.2 Motivation

Traditional microcontroller-based controllers execute instructions sequentially, which can introduce latency in time-sensitive operations. In contrast, FPGA-based implementation enables deterministic synchronous execution and true hardware parallelism, allowing multiple operations to occur simultaneously. This results in faster response time and improved system efficiency. Additionally, the reconfigurable architecture of FPGA devices provides flexibility for future modifications and feature enhancements, making them an attractive choice for modern embedded control system design.

2 Objectives

2.1 Primary Objective

To design and implement a hardware-based automatic washing machine controller using Finite State Machine (FSM) architecture on FPGA.

2.2 Specific Objectives

1. Develop a Moore FSM for washing cycle control.
2. Design memory registers for timing and state storage.
3. Implement ALU for arithmetic operations and comparisons.
4. Ensure safe operation using sensor validation.
5. Display operational states using LEDs and 7-segment display.

3 System Architecture

The proposed system architecture is composed of three major modules: the Control Unit based on a Moore Finite State Machine (FSM), Memory Registers, and an Arithmetic Logic Unit (ALU). These modules work together to ensure synchronized control, timing management, and arithmetic processing required for automatic washing machine operation.

3.1 Control Unit

The control unit serves as the core component of the system and is implemented using a Moore FSM architecture. It determines the current operational state of the washing machine and generates the corresponding output control signals. State transitions are managed based on external sensor inputs and timer values stored in memory registers. The Moore FSM structure ensures that outputs depend solely on the current state, providing stable and deterministic system behavior.

3.2 Memory Module

The memory module is responsible for storing all necessary state and timing information required for system operation. It maintains the current FSM state, wash timer, rinse timer, and spin timer values. Additionally, it stores the selected operating mode and the rinse counter used to track rinse cycle repetitions. These registers ensure synchronized timing control and proper sequencing of washing operations.

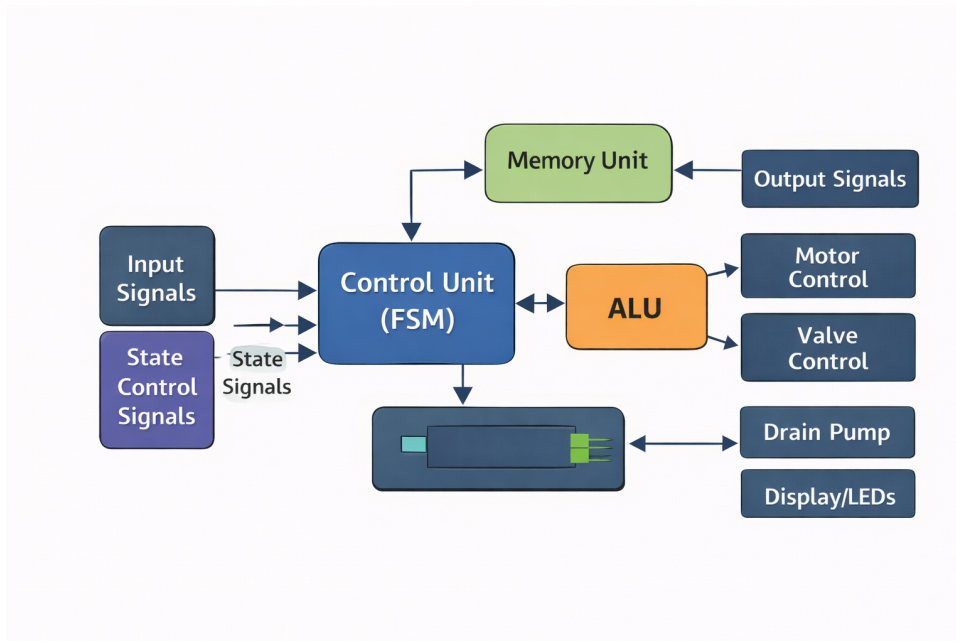


Figure 1. System Architecture

3.3 Arithmetic Logic Unit

The Arithmetic Logic Unit (ALU) performs the essential arithmetic and comparison operations required by the controller. It handles timer decrement operations, counter increment operations, and zero-detection comparisons that determine state transitions. By offloading arithmetic operations to the ALU, the system achieves efficient and organized hardware-level computation within the FPGA architecture.

4 Finite State Machine Design

The washing machine controller is implemented using a Moore Finite State Machine (FSM) architecture. In a Moore FSM, the outputs depend solely on the current state, ensuring stable and predictable system behavior. This approach simplifies hardware

implementation and guarantees deterministic state-based control within the FPGA environment.

4.1 FSM States

The controller consists of seven operational states: IDLE, FILL_WATER, WASH, RINSE, SPIN, DRAIN, and COMPLETE. The system begins in the IDLE state, waiting for user input. Upon activation, it sequentially transitions through water filling, washing, rinsing, spinning, and draining stages before reaching the COMPLETE state. Each state represents a distinct phase of the washing process and generates specific control outputs corresponding to that phase.

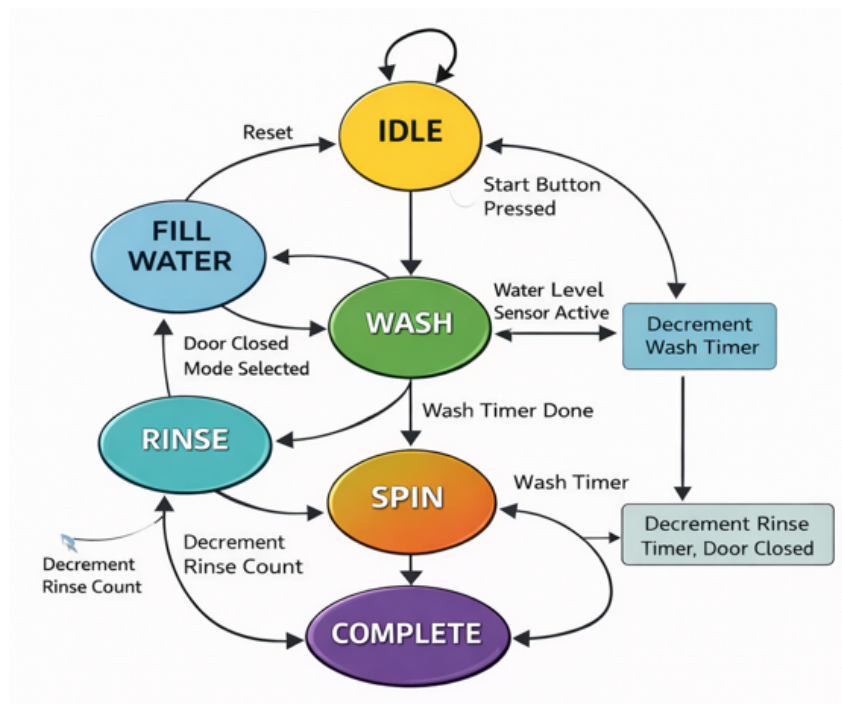


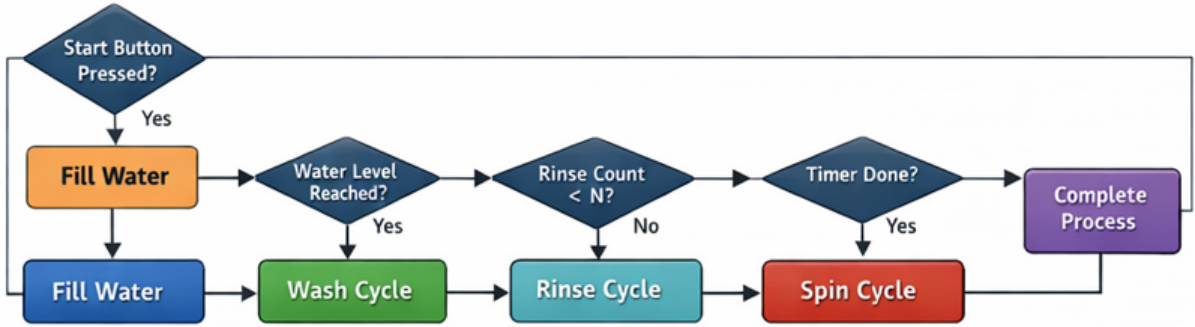
Figure 2. Working Flow Chart

4.2 State Transition Logic

State transitions are governed by predefined logical conditions based on input signals, sensor feedback, and timer values. The transition rules ensure that each operational phase is completed before moving to the next state. For example, the system transitions from IDLE to FILL_WATER when the start button is pressed, and from FILL_WATER to WASH when the water level sensor indicates a full condition. Timer-based transitions occur when the respective countdown value reaches zero, ensuring accurate timing control. The detailed transition conditions are summarized in Table 1.

Table 1. FSM Transition Conditions

Current State	Condition	Next State
IDLE	Start Pressed	FILL_WATER
FILL_WATER	Water Level = Full	WASH
WASH	Wash Timer = 0	RINSE
RINSE	Rinse Count Complete	SPIN
SPIN	Spin Timer = 0	DRAIN
DRAIN	Drain Timer = 0	COMPLETE
COMPLETE	Reset	IDLE

**Figure 3.** Working Flow Chart

5 Timing Model

The timing behavior of the washing machine controller is governed by a synchronous countdown mechanism implemented using timer registers. The timer value is decremented at each clock cycle according to the following update equation:

$$T_{n+1} = T_n - 1$$

This equation indicates that the next timer value is obtained by subtracting one from the current timer value at every clock pulse.

State transitions are controlled by the timer reaching zero. The transition condition can be expressed as:

$$\text{If } T = 0 \Rightarrow \text{Transition to Next State}$$

This condition ensures that the controller advances to the subsequent operational state only after the completion of the allocated time duration.

The overall state update mechanism of the finite state machine can be represented as:

$$State_{t+1} = f(State_t, Input, Timer)$$

where the next state depends on the current state, input signals, and the timer status.

This mathematical representation ensures deterministic and synchronized state progression within the FPGA-based control system.

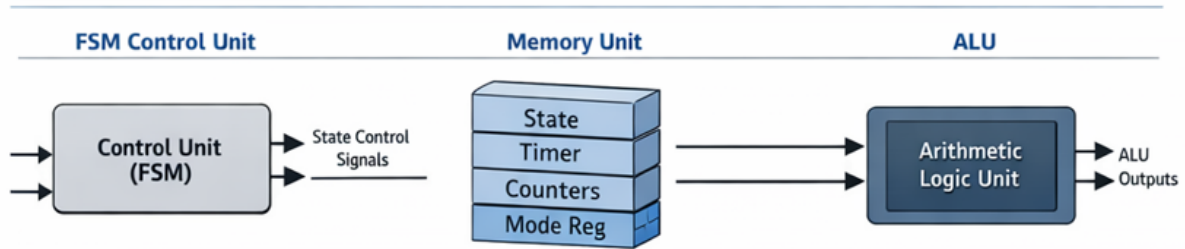


Figure 4. Key Component for Design

6 Mode Configuration

Table 2. Mode-Based Timing Configuration

Mode	Wash (s)	Rinse (s)	Spin (s)
Quick	5	3	4
Normal	10	5	6
Heavy	15	8	10

7 Input and Output Signals

The washing machine controller operates using several input and output signals to ensure proper functionality and safe operation. The input signals include the system clock, which synchronizes all sequential operations; a reset signal to initialize the system to the IDLE state; a start button to begin the washing cycle; a door sensor to ensure the door is securely closed before operation; a water level sensor to detect when the required water level has been reached; and a mode selector to choose between different washing modes.

The output signals control the hardware components of the system. These include the motor ON/OFF signal to activate or deactivate the drum motor, the motor direction control signal for rotational movement, the water inlet valve control for filling water, and the drain pump control for removing water. Additionally, status LEDs are used to indicate the current operational state, and a 7-segment display provides visual feedback of timing or state information.

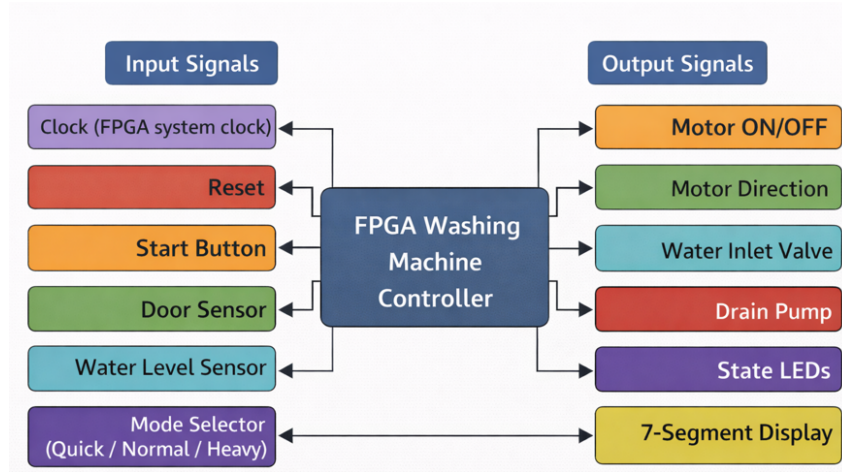


Figure 5. Input and Output

8 Safety Mechanisms

The washing machine controller incorporates several safety mechanisms to ensure reliable and secure operation. The motor is permitted to operate only when the door sensor confirms that the door is properly closed, thereby preventing accidental operation. Water filling is automatically stopped once the water level sensor detects that the required level has been reached, preventing overflow. The reset button is designed to immediately force the system back to the IDLE state, allowing safe reinitialization in case of unexpected behavior. Furthermore, all state transitions are strictly condition-controlled, ensuring that the system moves between operational stages only when predefined safety and timing conditions are satisfied.

9 Hardware Implementation

The proposed design was synthesized using Xilinx Vivado, targeting the Artix-7 FPGA available on the Basys 3 development board. The complete hardware development flow began with RTL design using Verilog to describe the system behavior. This was followed by functional simulation and verification to ensure correct operation of the FSM, timers, and control logic. After successful verification, the design underwent synthesis to translate the HDL code into gate-level representation, followed by implementation, which included placement and routing on the FPGA fabric. A bitstream file was then generated and programmed onto the FPGA board. Finally, hardware testing was conducted to validate real-time operation and verify correct functionality of the washing machine controller.

10 Testing and Verification

The system was thoroughly tested to ensure correct functional and hardware operation. Simulation was performed to verify proper FSM state transitions under different input conditions. The timer decrement behavior was examined to confirm accurate countdown and state progression. Mode selection functionality was tested to ensure correct timing configuration for different washing modes. Safety mechanisms were validated by checking door sensor conditions and reset functionality. Additionally, LED outputs and display

indicators were observed to confirm correct representation of operational states during real-time hardware testing.

11 Limitations and Future Work

Although the FPGA-based washing machine controller provides deterministic and reliable performance, it has certain limitations. The design requires knowledge of Hardware Description Languages (HDL), which may increase development complexity compared to traditional microcontroller-based systems. Additionally, FPGA-based implementations are generally more complex in terms of design flow, verification, and hardware resource management. The system is also constrained by the available FPGA resources, including logic cells, registers, and I/O pins, which may limit scalability for more advanced features.

Future improvements can enhance the functionality and versatility of the system. PWM-based motor speed control can be integrated to enable variable speed operation during different washing phases. An LCD interface may be added to provide more detailed user feedback and system status information. Furthermore, IoT-based monitoring can allow remote control and performance tracking through network connectivity. Integration of a temperature sensor would also enable automatic control of water temperature, making the system more advanced and energy-efficient.

12 Conclusion

The FPGA-based automatic washing machine controller demonstrates a structured hardware implementation of an embedded control system using FSM architecture. The integration of control logic, memory, and ALU ensures reliable, deterministic, and safe operation. FPGA implementation provides superior real-time performance compared to sequential microcontroller systems. The project successfully validates the applicability of FPGA-based control systems in modern appliance automation.

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